Alcohols, carboxylic acids, esters and polymers

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 2

Paper 2 only | Alcohols | [1]

Answer: B

Worked solution

"Prop-" means three carbon atoms and the "-ol" ending means the compound is an alcohol with an –OH group. The "1" shows the –OH is on the first carbon atom of the chain, so the structural formula is $CH_3CH_2CH_2OH$.

2 · Part 2

Paper 2 only | Alcohols | [1]

Answer: D

Worked solution

Learn the two ways of making ethanol separately. Ethene + steam: phosphoric acid catalyst, about 300 °C, about 60–70 atm. Fermentation of glucose: enzymes in yeast, about 30 °C, absence of air. Silica or alumina at 600–700 °C is used for cracking long-chain alkanes.

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3 · Part 2

Paper 2 only | Alcohols | [4]

Mark scheme

(a) **[3]**

- CH_3OH (1)
- butan-1-ol (1)

ALLOW butanol

- $C_4H_{10}O$ (1)

ALLOW C_4H_9OH

(b) **[1]**

- $C_5H_{12}O$ (1)

ALLOW $C_5H_{11}OH$

Worked solution

(a) Methanol has one carbon atom joined to the –OH group, so its structural formula is CH_3OH. $CH_3CH_2CH_2CH_2OH$ has four carbon atoms in a chain with –OH on the end carbon, so it is butan-1-ol (butanol is accepted). Counting the atoms in $CH_3CH_2CH_2CH_2OH$ gives 4 C, 10 H and 1 O, so the molecular formula is $C_4H_{10}O$.

(b) Methanol is CH_4O and butan-1-ol is $C_4H_{10}O$. Each extra carbon atom adds CH_2, so the pattern is CnH2n+2O. With five carbon atoms: $C_5H_{12}O$.

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4 · Part 2 Paper 2 only Alcohols [6]

Mark scheme

(a) **[1]**

- enzymes (1)

(b) **[3]**

- at 10 °C the reaction is too slow (1)

ALLOW enzymes are less active at 10 °C

- at 60 °C the enzymes are denatured OR destroyed OR stop working (1)

ALLOW yeast is killed at 60 °C

- 30 °C gives the fastest rate of fermentation without denaturing the enzymes (1)

ALLOW 30 °C is the optimum temperature for the enzymes

(c) **[2]**

- oxygen in the air would oxidise the ethanol (1)
- forming ethanoic acid, so less ethanol is produced (1)

ALLOW with air, yeast respires aerobically and makes carbon dioxide and water instead of ethanol, for either mark

Worked solution

(a) The biological catalysts in yeast are enzymes.

(b) Enzymes work fastest at their optimum temperature. At 10 °C the particles have less energy and the enzymes are less active, so fermentation is very slow. At 60 °C the enzymes are denatured (their shape is changed permanently), so they stop working and fermentation stops. About 30 °C gives a good rate while keeping the enzymes working.

(c) If air gets in, oxygen oxidises ethanol to ethanoic acid (this is how wine turns into vinegar), so the ethanol is lost. Keeping air out makes sure the glucose is converted into ethanol.

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5 · Part 2 Paper 2 only Alcohols [7]

Mark scheme

(a) **[3]**

- Mr of glucose = 180 and amount of glucose = 45.0 ÷ 180 = 0.250 mol (1)
- amount of ethanol = 2 × 0.250 = 0.500 mol (1)
- mass of ethanol = 0.500 × 46 = 23.0 g (1)

ALLOW 23 g

ECF from an incorrect Mr or amount

(b) **[2]**

- (18.4 ÷ 23.0) × 100 (1)
- 80.0% (1)

ALLOW 80%

ECF from the mass in (a)

(c) **[2]**

- amount of carbon dioxide = 2 × 0.250 = 0.500 mol (1)
- volume = 0.500 × 24 = 12.0 dm^3 (1)

ALLOW 12 dm^3

ECF from the amount of glucose in (a)

Worked solution

(a) Mr of glucose, $C_6H_{12}O_6$ = (6 × 12) + (12 × 1) + (6 × 16) = 180. Amount of glucose = 45.0 ÷ 180 = 0.250 mol. The equation shows 1 mol of glucose gives 2 mol of ethanol, so 0.500 mol of ethanol forms. Mr of ethanol, C_2H_5OH = (2 × 12) + (6 × 1) + 16 = 46. Mass = 0.500 × 46 = 23.0 g.

(b) Percentage yield = (actual yield ÷ theoretical yield) × 100 = (18.4 ÷ 23.0) × 100 = 80.0%.

(c) 1 mol of glucose also gives 2 mol of carbon dioxide, so 0.500 mol of CO_2 forms. Volume = 0.500 × 24 = 12.0 dm^3.

Check: 180 g of glucose would give 92 g of ethanol and 48 dm^3 of CO_2; 45.0 g is a quarter of 180 g, giving 23.0 g and 12.0 dm^3.

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6 · Part 2 Paper 2 only Alcohols [7]

Mark scheme

(a) **[2]**

- correct formulae of the missing substances: O_2, CO_2 and H_2O (1)
- correct balancing: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ (1)

ALLOW multiples

(b) **[4]**

- $Q = 100 \times 4.2 \times 30.0 = 12\,600$ J (1)

ALLOW 12.6 kJ

- mass of ethanol burned = 0.92 g and amount = $0.92 \div 46 = 0.0200$ mol (1)
- $12.6 \div 0.0200 = 630$ (1)
- $\Delta H = -630$ kJ/mol, with negative sign and unit (1)

ECF from an incorrect Q or amount

REJECT +630 kJ/mol for the final mark

(c) **[1]**

- heat is lost to the surroundings OR incomplete combustion of the ethanol (1)

ALLOW some heat warms the copper can OR some ethanol evaporates without burning

Worked solution

(a) Complete combustion of ethanol gives carbon dioxide and water. Balance carbon first ($2CO_2$), then hydrogen (6 H atoms make $3H_2O$), then oxygen: the products contain 4 + 3 = 7 O atoms, ethanol supplies 1, so 6 O atoms ($3O_2$) are needed. $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.

(b) Temperature rise = 49.5 − 19.5 = 30.0 °C. $Q = m \times c \times \Delta T = 100 \times 4.2 \times 30.0 = 12\,600$ J = 12.6 kJ. Mass of ethanol burned = 186.32 − 185.40 = 0.92 g. Mr of ethanol = 46, so amount = $0.92 \div 46 = 0.0200$ mol. $\Delta H = -12.6 \div 0.0200 = -630$ kJ/mol. The sign is negative because burning gives out heat (exothermic).

(c) Much of the heat from the flame goes into the air and the can rather than the water, and some ethanol may burn incompletely, so the measured heat is too small and ΔH is less negative.

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7 · Part 2 Paper 2 only Alcohols [5]

Mark scheme

(a) **[2]**

- potassium dichromate(VI) in dilute sulfuric acid (1)

ALLOW acidified potassium dichromate(VI)

- heat (1)

ALLOW warm OR heat under reflux

(b) **[2]**

- ethanol reacts with oxygen from the air (1)

ALLOW ethanol is oxidised by microbes

- forming ethanoic acid, which tastes sour (1)

ALLOW forming vinegar

(c) **[1]**

- it stops (fresh) air OR oxygen reaching the wine (1)

ALLOW less oxygen is available for the microbes OR for the oxidation

Worked solution

(a) In the laboratory, ethanol is heated with potassium dichromate(VI) in dilute sulfuric acid. This oxidising agent converts ethanol into ethanoic acid.

(b) Microbes in the open wine use oxygen from the air to oxidise ethanol into ethanoic acid (microbial oxidation). Ethanoic acid is the acid in vinegar, so the wine tastes sour.

(c) A tight seal keeps air out. Without a supply of oxygen, the microbial oxidation of ethanol slows down.

8 · Part 2 Paper 2 only Carboxylic acids [1]

Answer: A

Worked solution

Count every carbon atom, including the one in the –COOH group: CH_3, CH_2, CH_2, COOH gives four. Four carbon atoms is "but-" and the –COOH group gives the ending "-oic acid", so the name is butanoic acid.

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9 · Part 2 Paper 2 only Carboxylic acids [1]

Answer: C

Worked solution

Vinegar is mainly water with ethanoic acid, CH_3COOH, dissolved in it. It forms when ethanol in wine is oxidised by microbes.

10 · Part 2 Paper 2 only Carboxylic acids [5]

Mark scheme

(a) **[2]**

- methanoic acid (1)
- CH_3CH_2COOH (1)

ALLOW C_2H_5COOH

(b) **[2]**

- both contain the same functional group, –COOH (1)

ALLOW carboxylic acid group OR –C(=O)–OH

- they have the same general formula OR similar chemical properties OR they differ by CH_2 units (1)

ALLOW show a trend in physical properties

(c) **[1]**

- pentanoic acid (1)

Worked solution

(a) HCOOH has one carbon atom (the carbon in –COOH), so it is methanoic acid. Propanoic acid has three carbon atoms: CH_3CH_2COOH.

(b) Members of a homologous series have the same functional group, here –COOH, so they react in similar ways. They also fit the same general formula, and each member differs from the next by CH_2.

(c) Five carbon atoms is "pent-", so the acid is pentanoic acid.

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11 · Part 2

Paper 2 only | Carboxylic acids | [7]

Mark scheme

(a) **[2]**

- fizzing OR bubbles OR effervescence (1)
- the solid gets smaller OR disappears OR dissolves (1)

IGNORE "a gas is produced" (not an observation)

REJECT "carbon dioxide is produced" as an observation

(b) **[2]**

- 2 in front of CH_3COOH (1)
- H_2O and CO_2 as the two products (1)

(c) **[3]**

- amount of calcium carbonate = 0.0400 ÷ 2 = 0.0200 mol (1)
- Mr of $CaCO_3$ = 100 (1)
- mass = 0.0200 × 100 = 2.00 g (1)

ALLOW 2 g

ECF from the mole ratio in (b) OR an incorrect Mr

Worked solution

(a) The acid reacts with the carbonate to make carbon dioxide gas, so you see fizzing. The calcium carbonate is used up, so the solid disappears and a solution forms.

(b) Acid + carbonate → salt + water + carbon dioxide. Each formula unit of $(CH_3COO)_2Ca$ contains two ethanoate ions, so two molecules of ethanoic acid are needed: $CaCO_3 + 2CH_3COOH \rightarrow (CH_3COO)_2Ca + H_2O + CO_2$.

(c) The equation shows 2 mol of ethanoic acid react with 1 mol of calcium carbonate. Amount of $CaCO_3$ = 0.0400 ÷ 2 = 0.0200 mol. Mr of $CaCO_3$ = 40 + 12 + (3 × 16) = 100. Mass = 0.0200 × 100 = 2.00 g.

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12 · Part 2 Paper 2 only Carboxylic acids [6]

Mark scheme

(a) **[1]**

- zinc ethanoate (1)

(b) **[3]**

- amount of zinc = 0.975 ÷ 65 = 0.0150 mol (1)
- amount of hydrogen = 0.0150 mol (1)
- volume = 0.0150 × 24 000 = 360 cm^3 (1)

ALLOW 0.360 dm^3 for 2 marks only (unit not converted to cm^3)

ECF from an incorrect amount

(c) **[2]**

- put a lighted splint into (the mouth of a tube of) the gas (1)
- squeaky pop (1)

REJECT glowing splint

Worked solution

(a) The metal gives the first part of the name and ethanoic acid gives "ethanoate", so the salt is zinc ethanoate.

(b) Ar of zinc = 65, so amount of zinc = 0.975 ÷ 65 = 0.0150 mol. The equation shows 1 mol of zinc gives 1 mol of hydrogen, so 0.0150 mol of H_2 forms. 1 mol of gas occupies 24 dm^3 = 24 000 cm^3, so volume = 0.0150 × 24 000 = 360 cm^3.

(c) Hold a lighted splint at the mouth of a test tube of the gas. Hydrogen burns with a squeaky pop.

13 · Part 2 Paper 2 only Esters [1]

Answer: B

Worked solution

Look for the functional group. In an ester, the –COO– group has a carbon chain on both sides. In a carboxylic acid the group ends in –COOH, with a hydrogen atom on the oxygen.

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14 · Part 2 Paper 2 only Esters [1]

Answer: A

Worked solution

Butanoic acid is $CH_3CH_2CH_2COOH$. In the ester, the H of the –COOH group is replaced by the alkyl group from the alcohol, which is CH_3 from methanol. This gives $CH_3CH_2CH_2COOCH_3$. The name puts the alcohol part first: methyl butanoate.

15 · Part 2 Paper 2 only Esters [1]

Answer: D

Worked solution

Each homologous series has its own functional group: alkenes >C=C<, alcohols –OH, carboxylic acids –COOH and esters –COO–. In an ester, the oxygen atom that would carry an H in a carboxylic acid is joined to a carbon chain instead.

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16 · Part 2

Paper 2 only | Esters | [5]

Mark scheme

(a) **[2]**

- ethanol (1)
- ethanoic acid (1)

(b) **[1]**

- $CH_3COOCH_2CH_3$ (1)

ALLOW $CH_3COOC_2H_5$ OR $CH_3CO_2CH_2CH_3$

(c) **[1]**

- water (1)

(d) **[1]**

- $C_4H_8O_2$ (1)

Worked solution

(a) The name of an ester comes from its alcohol ("ethyl" from ethanol) and its carboxylic acid ("ethanoate" from ethanoic acid).

(b) Ethanoic acid is CH_3COOH. The H of the –COOH group is replaced by the ethyl group, CH_2CH_3, from ethanol, giving $CH_3COOCH_2CH_3$.

(c) ethanol + ethanoic acid → ethyl ethanoate + water.

(d) Counting atoms in $CH_3COOCH_2CH_3$: 4 C, 8 H and 2 O, so the molecular formula is $C_4H_8O_2$.

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17 · Part 2

Paper 2 only | Esters | [6]

Mark scheme

(a) **[1]**

- catalyst (1)

ALLOW speeds up the reaction

(b) **[2]**

- propan-1-ol OR ethanoic acid OR the ester is flammable (1)
- a flame could set fire to the vapour OR the water bath has no naked flame (1)

ALLOW the water bath heats the mixture gently and evenly, for the second mark

(c) **[2]**

- sodium carbonate neutralises (unreacted) acid (1)

ALLOW neutralises the ethanoic acid OR the sulfuric acid

- so the smell of the acid does not hide the smell of the ester (1)

ALLOW the ester forms a separate layer on top of the solution OR so the ester can be separated

(d) **[1]**

- a distinctive OR fruity OR sweet smell (1)

ALLOW an oily OR insoluble layer forms on top of the solution

IGNORE "it smells" without a description

Worked solution

(a) Concentrated sulfuric acid is the acid catalyst for the reaction between an alcohol and a carboxylic acid.

(b) The alcohol and the ester are flammable and give off vapour when warm. A naked flame could ignite the vapour, so a beaker of hot water is used to warm the tube safely and gently.

(c) The mixture still contains unreacted ethanoic acid and the sulfuric acid. Sodium carbonate neutralises these acids, so the sharp vinegar smell of ethanoic acid does not hide the ester. The ester does not mix with water and floats as a separate layer.

(d) Esters have distinctive, often fruity, smells, so a new sweet or fruity smell shows an ester has formed.

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18 · Part 2

Paper 2 only | Esters | [7]

Mark scheme

(a) **[5]**

- methyl ethanoate (1)
- CH_3COOCH_3 (1)
- ethanol (1)
- propanoic acid (1)
- $CH_3CH_2COOCH_2CH_3$ (1)

ALLOW $C_2H_5COOC_2H_5$ OR $CH_3CH_2CO_2CH_2CH_3$ for the last mark

ECF structural formula correct for an incorrect name given in the same row

(b) **[2]**

- butan-1-ol (1)

ALLOW butanol

- ethanoic acid (1)

Worked solution

(a) An ester is named from its alcohol (first part, ending "-yl") and its acid (second part, ending "-oate"). Methanol and ethanoic acid give methyl ethanoate; the H of –COOH in CH_3COOH is replaced by CH_3, giving CH_3COOCH_3. Ethyl propanoate comes from ethanol ("ethyl") and propanoic acid ("propanoate"); replacing the H of CH_3CH_2COOH with CH_2CH_3 gives $CH_3CH_2COOCH_2CH_3$.

(b) Split the ester at the –COO– group. The part ending in COO is CH_3COO–, which comes from ethanoic acid, CH_3COOH. The part joined to the single-bonded oxygen is $CH_2CH_2CH_2CH_3$, a four-carbon chain, which comes from butan-1-ol, $CH_3CH_2CH_2CH_2OH$. The ester is butyl ethanoate.

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19 · Part 2

Paper 2 only | Esters | [7]

Mark scheme

(a) **[1]**

- $CH_3CH_2CH_2COOCH_2CH_3$ (1)

ALLOW $C_3H_7COOC_2H_5$ OR $CH_3CH_2CH_2CO_2CH_2CH_3$

(b) **[1]**

- (6 × 12) + (12 × 1) + (2 × 16) = 116 (1)

(c) **[2]**

- amount of butanoic acid = 8.80 ÷ 88 = 0.100 mol (1)
- mass of ethyl butanoate = 0.100 × 116 = 11.6 g (1)

ECF from the Mr in (b)

(d) **[2]**

- (8.12 ÷ 11.6) × 100 (1)
- 70.0% (1)

ALLOW 70%

ECF from the mass in (c)

(e) **[1]**

- the reaction is reversible OR does not go to completion (1)

ALLOW some product is lost when the ester is separated OR some ester evaporates OR side reactions OR not all the acid reacts

Worked solution

(a) Butanoic acid is $CH_3CH_2CH_2COOH$. The H of the –COOH group is replaced by the ethyl group from ethanol, giving $CH_3CH_2CH_2COOCH_2CH_3$.

(b) The molecular formula is $C_6H_{12}O_2$, so Mr = (6 × 12) + (12 × 1) + (2 × 16) = 72 + 12 + 32 = 116.

(c) Amount of butanoic acid = 8.80 ÷ 88 = 0.100 mol. This forms 0.100 mol of ester. Mass = 0.100 × 116 = 11.6 g.

(d) Percentage yield = (8.12 ÷ 11.6) × 100 = 70.0%.

(e) The reaction between an alcohol and a carboxylic acid is reversible, so it never fully turns into products. Some ester is also lost when it is separated from the mixture, and esters are volatile, so some evaporates.

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20 · Part 2 Paper 2 only Esters [5]

Mark scheme

(a) **[2]**

- volatile OR evaporate easily (1)
- distinctive OR pleasant smell (1)

ALLOW fruity smell

(b) **[2]**

- ethyl methanoate (1)
- it has the lowest boiling point (1)
- no mark for the reason if the wrong ester is chosen

(c) **[1]**

- (food) flavourings (1)

ALLOW to give food a fruity flavour

Worked solution

(a) A perfume must release its smell into the air. Esters are volatile, so they evaporate easily at body temperature, and they have distinctive, pleasant smells.

(b) The ester with the lowest boiling point needs the least energy to turn into a gas, so it evaporates most easily. Ethyl methanoate boils at 54 °C, lower than the other two.

(c) Because of their fruity smells and tastes, esters are used as food flavourings.

21 · Part 2 Paper 1 & 2 Synthetic polymers [1]

Answer: C

Worked solution

Addition polymers are made from monomers containing a C=C double bond. In polymerisation, one bond of the double bond breaks and the carbon atoms join to neighbouring monomer molecules, building a long chain.

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22 · Part 2 Paper 1 & 2 Synthetic polymers [1]

Answer: A

Worked solution

To find the monomer, take the two carbon atoms in the brackets, remove the extension bonds and put a double bond between them. Keep every atom attached to each carbon exactly the same. CH_2 and CCl_2 joined by a double bond give $CH_2{=}CCl_2$.

23 · Part 2 Paper 2 only Synthetic polymers [1]

Answer: B

Worked solution

Each time a –COOH group reacts with an –OH group, an ester link forms and a small molecule of water is given off. Repeating this many times gives a long polyester chain plus many water molecules.

24 · Part 2 Paper 1 & 2 Synthetic polymers [5]

Mark scheme

(a) **[3]**

- $-[CH_2{-}CH_2]-n$ (1)
- $CH_2{=}CHCl$ (1)
- $-[CF_2{-}CF_2]-n$ (1)

ALLOW equivalent formats such as $[-CH_2{-}CH_2-]n$

(b) **[2]**

- ethane does not have a carbon–carbon double bond OR ethane is saturated (1)
- addition polymerisation needs the C=C double bond to open so that monomers can join together (1)

Worked solution

(a) For each monomer, the C=C double bond becomes a single bond and each carbon gets an extension bond to join the next unit: ethene gives $-[CH_2{-}CH_2]-n$ and tetrafluoroethene gives $-[CF_2{-}CF_2]-n$. Working backwards from $-[CH_2{-}CHCl]-n$, putting a double bond between the two carbons gives chloroethene, $CH_2{=}CHCl$.

(b) Ethane has only single bonds, so there is no double bond that can open up. Without this, the molecules have no way of joining together in a chain by addition.

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25 · Part 2 Paper 1 & 2 Synthetic polymers [5]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- poly(chloroethene) is inert OR unreactive (1)
- it is not biodegradable OR is not broken down by microorganisms (1)
- so it stays in landfill sites for a very long time OR landfill sites fill up (1)
- burning produces toxic gases (1)
- hydrogen chloride is produced because the polymer contains chlorine (1)

ALLOW acidic gases

- carbon monoxide is produced by incomplete combustion (1)

IGNORE "pollution" without explanation

(b) **[1]**

- recycle it OR reuse the items (1)

ALLOW melt it down to make new items OR use less of it by replacing it with a biodegradable polymer

Worked solution

(a) Addition polymers such as poly(chloroethene) are very unreactive. Microorganisms cannot break them down, so they are not biodegradable and remain in landfill for hundreds of years, using up space. Burning them gets rid of the waste, but poly(chloroethene) contains chlorine, so burning releases toxic hydrogen chloride gas. Incomplete combustion also produces toxic carbon monoxide.

(b) Recycling (or reusing) the items avoids both landfill and burning.

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26 · Part 2 Paper 1 & 2 Synthetic polymers [6]

Mark scheme

(a) **[1]**

- –[CH_2–$CH(CH_3)$]–n (1)

ALLOW [–CH_2–$CH(CH_3)$–]n OR –[$CH(CH_3)$–CH_2]–n

(b) **[1]**

- (3 × 12) + (6 × 1) = 42 (1)

(c) **[2]**

- 126 000 ÷ 42 (1)
- 3000 (1)

ECF from the Mr in (b)

(d) **[2]**

- in addition polymerisation the double bond opens and the monomers simply join together (1)
- no atoms are lost OR no other product is formed, so the repeat unit contains the same atoms as the monomer (1)

ALLOW the polymer is the only product

Worked solution

(a) Propene is CH_2=$CHCH_3$. The double bond opens and the CH_3 group becomes a side group on the chain, giving –[CH_2–$CH(CH_3)$]–n.

(b) Propene, C_3H_6: Mr = (3 × 12) + (6 × 1) = 42.

(c) Each propene molecule adds 42 to the Mr of the chain, so the number of molecules = 126 000 ÷ 42 = 3000.

(d) In addition polymerisation nothing is removed from the monomer: the C=C bond becomes a single bond and the molecules link up. The repeat unit therefore has exactly the same atoms as propene (C_3H_6), so its Mr is also 42.

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27 · Part 2 Paper 1 & 2 Synthetic polymers [5]

Mark scheme

(a) **[1]**

- CF_2=CFCl (1)

(b) **[1]**

- –[CH(CH_3)–CH(CH_3)]–n (1)

ALLOW [–CH(CH_3)–CH(CH_3)–]n

(c) **[2]**

- repeat unit –[CH_2–CHF]–n (1)
- monomer CH_2=CHF (1)

(d) **[1]**

- a small molecule that joins with many others to form a polymer (1)

Worked solution

(a) Put a double bond between the two carbons of the repeat unit and keep the same atoms on each: CF_2=CFCl.

(b) In but-2-ene, CH_3CH=$CHCH_3$, the double bond is between the two middle carbons. Each of these carbons carries a CH_3 group, so the repeat unit is –[CH(CH_3)–CH(CH_3)]–n.

(c) The chain repeats the pattern CH_2–CHF, so the repeat unit is –[CH_2–CHF]–n. Putting a double bond between the two carbons gives the monomer CH_2=CHF.

(d) A monomer is one of the many small molecules that join together to make a polymer.

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28 · Part 2 Paper 2 only Synthetic polymers [5]

Mark scheme

(a) **[1]**

- condensation (polymerisation) (1)

(b) **[2]**

- ester links –CO–O– between the acid part and the diol part (1)
- complete repeat unit –[CO–CO–O–CH_2CH_2–O]–n (1)

ALLOW –[OC–COO–CH_2CH_2–O]–n OR –[O–CH_2CH_2–O–CO–CO]–n OR [–C(=O)–C(=O)–O–CH_2CH_2–O–]n

(c) **[2]**

- each monomer can react at both ends of its molecule (1)
- so it can join to two other molecules and the chain can keep growing (1)

ALLOW with only one functional group, the reaction would stop after one link OR would form a single ester molecule

Worked solution

(a) The –COOH groups react with –OH groups and a small molecule (water) is given off each time a link forms, so this is condensation polymerisation.

(b) Each –COOH group loses its OH and each –OH group loses its H; these leave as water. The remaining parts join through ester links. One acid unit (–CO–CO–) and one diol unit (–O–CH_2CH_2–O–) together make the repeat unit –[CO–CO–O–CH_2CH_2–O]–n, with 2n H_2O also formed.

(c) A molecule with a functional group at each end can link to one molecule on its left and another on its right. That lets the chain keep growing. If each molecule had only one functional group, two molecules would join and there would be nothing left to react, giving a simple ester instead of a polymer.

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29 · Part 2

Paper 2 only | Synthetic polymers | [5]

Mark scheme

(a) **[1]**

- –[CH_2–$C(CH_3)_2$]–n (1)

ALLOW [–CH_2–$C(CH_3)_2$–]n

(b) **[2]**

- ester links –CO–O– between the acid part and the diol part (1)
- complete repeat unit –[CO–CH_2CH_2–CO–O–$CH_2CH_2CH_2$–O]–n (1)

ALLOW –[O–$CH_2CH_2CH_2$–O–CO–CH_2CH_2–CO]–n OR [–C(=O)–CH_2CH_2–C(=O)–O–$CH_2CH_2CH_2$–O–]n

(c) **[2]**

- polymer S is biodegradable OR can be broken down by microorganisms (1)
- polymer R is inert OR not biodegradable, so it stays in the landfill site for a very long time (1)

Worked solution

(a) The double bond in CH_2=$C(CH_3)_2$ opens. One carbon carries two H atoms and the other carries two CH_3 groups, giving –[CH_2–$C(CH_3)_2$]–n.

(b) The –COOH groups of the acid react with the –OH groups of the diol, losing water. The acid part –CO–CH_2CH_2–CO– is joined through ester links to the diol part –O–$CH_2CH_2CH_2$–O–, so the repeat unit is –[CO–CH_2CH_2–CO–O–$CH_2CH_2CH_2$–O]–n.

(c) Biopolyesters are biodegradable: microorganisms break them down over time, so they do not build up in landfill. Addition polymers like R are unreactive and are not broken down, so they stay in landfill for hundreds of years.

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Mark scheme

(a) **[5]**

Indicative points, any five (1 each):

- poly(ethene) forms by addition polymerisation (1)
- the polyester forms by condensation polymerisation (1)
- the ethene monomer contains a C=C double bond, which opens (1)
- the polyester monomers each have two functional groups, –COOH and –OH (1)
- poly(ethene) is made from one type of monomer but a polyester is made from two different monomers, a dicarboxylic acid and a diol (1)
- addition polymerisation gives the polymer as the only product (1)
- condensation polymerisation also gives water (a small molecule) (1)

(b) **[1]**

- both join many small monomer molecules to form long-chain molecules (1)

ALLOW both form polymers OR both produce large molecules with a repeat unit

Worked solution

(a) Poly(ethene) is made by addition polymerisation. Ethene has a C=C double bond; one bond of it opens and thousands of ethene molecules join. Only one type of monomer is needed and the polymer is the only product. A polyester is made by condensation polymerisation. Two different monomers are used, a dicarboxylic acid (two –COOH groups) and a diol (two –OH groups). Each time a –COOH group reacts with an –OH group, an ester link forms and a molecule of water is given off, so there are two products: the polymer and water.

(b) In both types, many small monomer molecules join together to make very long chain molecules (polymers) built from a repeat unit.

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